# 網路架構
## ONTAP 9

NetApp
May 10, 2022

This PDF was generated from https://docs.netapp.com/zh-tw/ontap/concepts/network-connectivity-concept.html on May 10, 2022. Always check docs.netapp.com for the latest.

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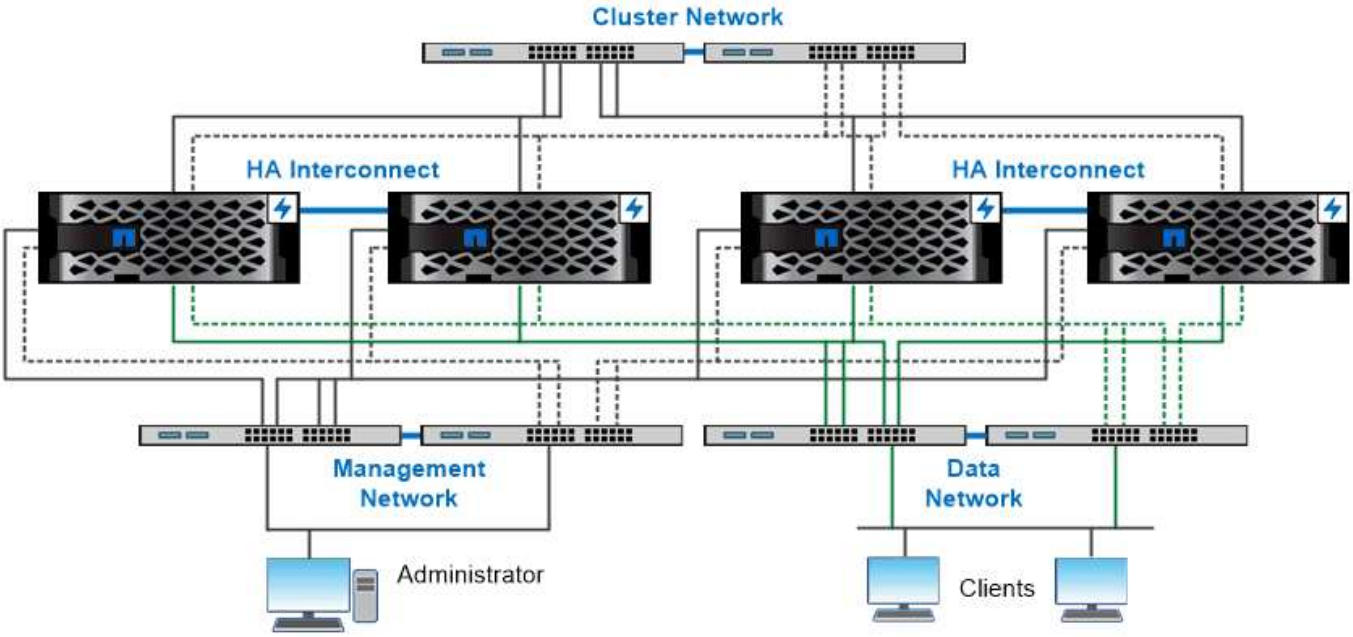
# 網路架構

## 網路架構總覽

實作資料中心的網路架構ONTAP 通常包含叢集互連、叢集管理的管理網路、以及資料網路。NIC（網路介面卡）提供實體連接埠以供乙太網路連線使用。HBA（主機匯流排介面卡）提供實體連接埠以供FC連線使用。




*The network architecture for an ONTAP datacenter implementation typically consists of a cluster interconnect, a management network for cluster administration, and a data network.*

## 邏輯連接埠

除了每個節點上提供的實體連接埠之外、您也可以使用_logical連接埠_來管理網路流量。邏輯連接埠是介面群組或VLAN。

### 介面群組

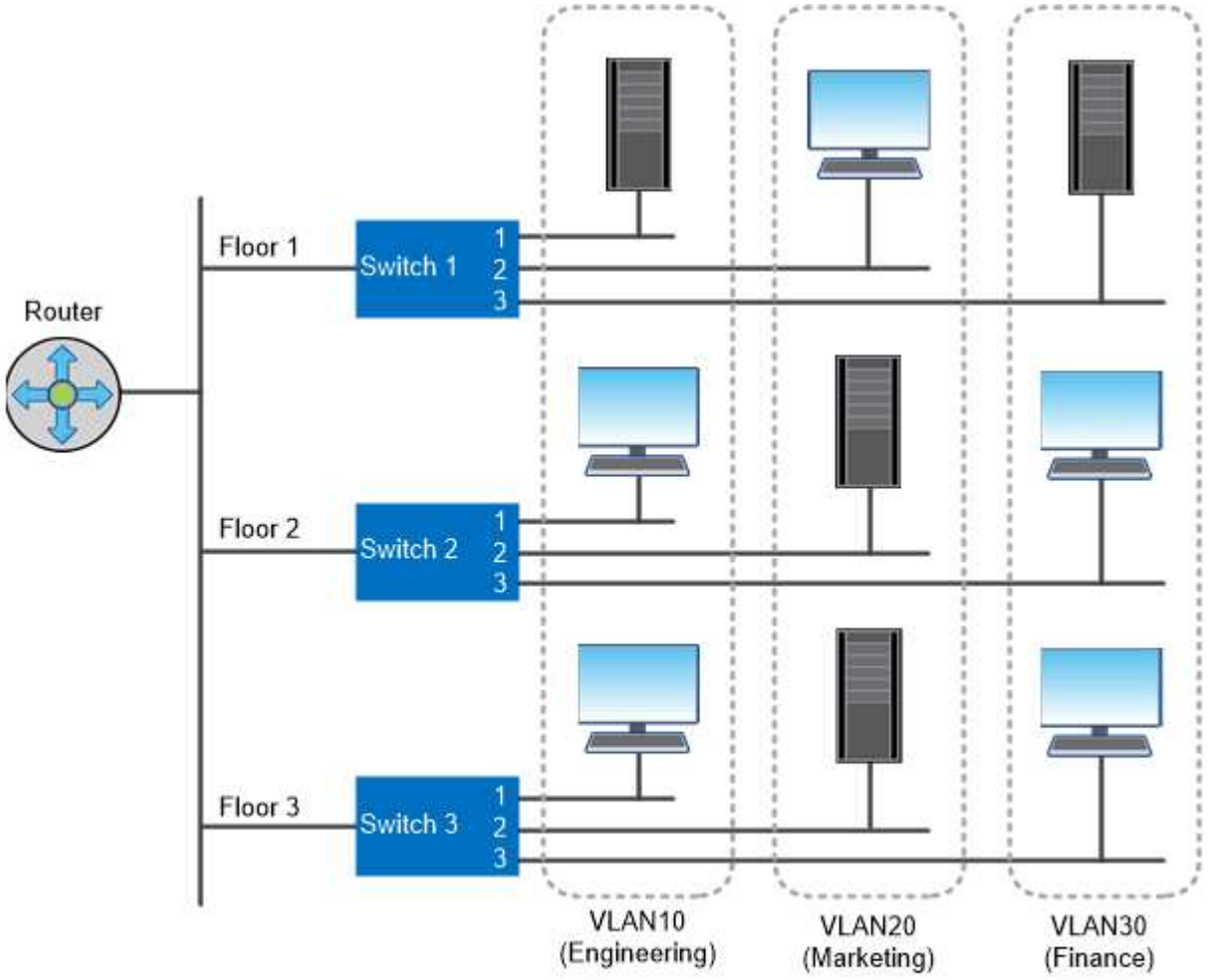
_Interface Groups_將多個實體連接埠合併成單一邏輯「'trunk port.'」 您可能想要在不同的PCI插槽中建立一個介面群組、由NIC的連接埠組成、以確保不會發生插槽故障而導致業務關鍵流量下降。

介面群組可以是單一模式、多重模式或動態多重模式。每種模式都提供不同的容錯層級。您可以使用任一類型的多重模式介面群組來平衡網路流量負載。

### VLAN

_vlans_可將流量從網路連接埠（可能是介面群組）分隔成以交換器連接埠為基礎而非實體邊界所定義的邏輯區段。屬於VLAN的_end-stations是根據功能或應用程式而定。

您可以依部門（例如工程與行銷部門）或專案（例如releasease1和releasease2）來分組終端站台。由於終端站台的實體鄰近與VLAN無關、因此終端站台可在地理上處於遠端位置。




*You can use VLANs to segregate traffic by department.*

# 支援業界標準網路技術

## 支援所有主要的產業標準網路技術。ONTAP關鍵技術包括IPspaces、DNS負載平衡及SNMP設陷。

廣播網域、容錯移轉群組及子網路的說明請參閱 NAS路徑容錯移轉。

### IPspaces

您可以使用_IPspace_為叢集中的每個虛擬資料伺服器建立不同的IP位址空間。這樣做可讓管理性分隔網路網域中的用戶端存取叢集資料、同時使用相同IP位址子網路範圍中重疊的IP位址。

例如、服務供應商可以為使用相同IP位址的租戶設定不同的IPspace、以存取叢集。

### DNS負載平衡

您可以使用_DNS負載平衡_、在可用的連接埠之間分散使用者網路流量。DNS伺服器會根據介面上掛載的用戶端數目、動態選取流量的網路介面。

### SNMP設陷

您可以使用_SNMP陷阱_定期檢查作業臨界值或故障。SNMP設陷會擷取從SNMP代理程式非同步傳送至SNMP管理程式的系統監控資訊。

- FIPS相容性*

所有SSL連線均符合聯邦資訊處理標準（FIPS）140-2。ONTAP您可以開啟和關閉SSL FIPS模式、全域設定SSL傳輸協定、以及關閉任何弱式加密器、例如RC4。

## RDMA總覽

如果您有對延遲敏感或高頻寬的工作負載、您可能會想要善用ONTAP的讀取直接記憶體存取（RDMA）產品。RDMA可讓資料直接複製到儲存系統記憶體和主機系統記憶體之間、避免CPU中斷和負荷。

從功能組態9.10.1開始ONTAP 、您就可以進行設定 "NFS over RDMA" 在支援NVIDIA GPU的主機上、啟用NVIDIA GPUDirect Storage以GPU加速工作負載。

此外、針對採用Pensando叢集NIC的A400或A400儲存系統的非功能性使用者、推出RDMA叢集互連技術。ONTAP ONTAPRDMA叢集互連可減少延遲、縮短容錯移轉時間、並加速叢集中節點之間的通訊。如果已設定適當的儲存系統、則不需要額外的組態。

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